

Network Configuration Lab Report

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1. Lab Objectives

Configure a small enterprise network with VLANs, inter-VLAN routing, DHCP, and basic ACLs using Cisco Packet Tracer.

2. Network Topology

The network consists of: 1x Layer 3 Switch (Core), 2x Layer 2 Switches (Access), 1x Router (Gateway), 3x VLANs (Management=10, Staff=20, Guest=30), DHCP Server on VLAN 10.

3. Configuration Commands

3.1 VLAN Configuration (Core Switch)

```
Switch(config)# vlan 10
Switch(config-vlan)# name Management
Switch(config)# vlan 20
Switch(config-vlan)# name Staff
Switch(config)# vlan 30
Switch(config-vlan)# name Guest
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config)# interface vlan 30
Switch(config-if)# ip address 192.168.30.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config)# ip routing
```

3.2 DHCP Configuration

```
Router(config)# ip dhcp pool STAFF
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(config)# ip dhcp pool GUEST
Router(dhcp-config)# network 192.168.30.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.30.1
Router(dhcp-config)# lease 0 4
```

3.3 ACL Configuration

```
Router(config)# access-list 100 deny ip 192.168.30.0 0.0.0.255 192.168.10.0
0.0.0.255
Router(config)# access-list 100 permit ip any any
Router(config)# interface vlan 30
Router(config-if)# ip access-group 100 in
```

4. Verification Results

- Ping from VLAN 20 to VLAN 10: SUCCESS (avg 2ms)
- Ping from VLAN 30 to VLAN 10: BLOCKED (ACL working)
- DHCP lease on VLAN 20: 192.168.20.101 assigned
- Inter-VLAN routing: All permitted paths working

- Guest isolation: Confirmed no access to management VLAN

5. Conclusion

Successfully configured enterprise network with VLAN segmentation, inter-VLAN routing via L3 switch, DHCP for dynamic addressing, and ACLs for guest network isolation. All verification tests passed.